

KodetoCareer — Database Schema

Version 1.0 | PostgreSQL 15

1. Schema Overview

The database uses a single PostgreSQL schema (public) with multi-tenancy enforced at the application layer through college_id scoping on all tenant-specific tables. Row-Level Security (RLS) is used for the college admin role.

Naming Conventions:

- Table names: snake_case, plural
- Column names: snake_case
- Primary keys: id (UUID)
- Foreign keys: {table_singular}_id
- Timestamps: created_at, updated_at on all tables
- Soft deletes: deleted_at (nullable) on key tables

2. Entity Relationship Overview



```

        ■■< trainers
        ■■< college_admins
        ■■< super_admins

notifications ■■< notification_recipients
jobs ■■< job_interests

---

```

3. Complete Table Definitions

3.1 Core User Tables

```

-- =====
-- USERS (Base authentication table for all user types)
-- =====
CREATE TABLE users (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    email             VARCHAR(255) UNIQUE NOT NULL,
    password_hash     VARCHAR(255) NOT NULL,
    role              VARCHAR(50) NOT NULL CHECK (role IN ('SUPER_ADMIN', 'COLLEGE_ADMIN'...
    first_name        VARCHAR(100) NOT NULL,
    last_name         VARCHAR(100) NOT NULL,
    phone             VARCHAR(20),
    avatar_url        TEXT,
    is_active          BOOLEAN DEFAULT true NOT NULL,
    is_email_verified BOOLEAN DEFAULT false NOT NULL,
    last_login_at     TIMESTAMPTZ,
    created_at        TIMESTAMPTZ DEFAULT NOW() NOT NULL,
    updated_at        TIMESTAMPTZ DEFAULT NOW() NOT NULL,
    deleted_at        TIMESTAMPTZ
);

CREATE INDEX idx_users_email ON users(email);
CREATE INDEX idx_users_role ON users(role);
CREATE INDEX idx_users_active ON users(is_active) WHERE is_active = true;

-- =====
-- EMAIL VERIFICATIONS
-- =====
CREATE TABLE email_verifications (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id           UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
    otp               VARCHAR(6) NOT NULL,
    type              VARCHAR(50) NOT NULL CHECK (type IN ('EMAIL_VERIFY', 'PASSWORD_RESET')),
    expires_at        TIMESTAMPTZ NOT NULL,
    used_at           TIMESTAMPTZ,
    created_at        TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

CREATE INDEX idx_email_verifications_user ON email_verifications(user_id);

```

```

-- =====
-- REFRESH TOKENS
-- =====
CREATE TABLE refresh_tokens (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id     UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
    token_hash  VARCHAR(255) UNIQUE NOT NULL,
    device_info JSONB,
    expires_at  TIMESTAMPTZ NOT NULL,
    revoked_at  TIMESTAMPTZ,
    created_at  TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

CREATE INDEX idx_refresh_tokens_user ON refresh_tokens(user_id);
CREATE INDEX idx_refresh_tokens_hash ON refresh_tokens(token_hash);

```

```
---
```

3.2 Organisation Tables

```

-- =====
-- COLLEGES
-- =====
CREATE TABLE colleges (
    id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    name        VARCHAR(255) NOT NULL,
    code        VARCHAR(20) UNIQUE NOT NULL,          -- e.g., 'JNITU', 'LNCT'
    address     TEXT,
    city        VARCHAR(100),
    state       VARCHAR(100),
    pincode     VARCHAR(10),
    contact_name VARCHAR(200),
    contact_email VARCHAR(255),
    contact_phone VARCHAR(20),
    logo_url    TEXT,
    website_url TEXT,
    is_active   BOOLEAN DEFAULT true NOT NULL,
    onboarded_at DATE,
    contract_start DATE,
    contract_end DATE,
    notes       TEXT,
    metadata    JSONB DEFAULT '{}',
    created_by  UUID REFERENCES users(id),
    created_at  TIMESTAMPTZ DEFAULT NOW() NOT NULL,
    updated_at  TIMESTAMPTZ DEFAULT NOW() NOT NULL,
    deleted_at  TIMESTAMPTZ
);

CREATE INDEX idx_colleges_code ON colleges(code);
CREATE INDEX idx_colleges_active ON colleges(is_active) WHERE is_active = true;

-- =====
-- COLLEGE ADMINS (links a USER to a COLLEGE with admin role)
-- =====

```

```

CREATE TABLE college_admins (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id     UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
  college_id  UUID NOT NULL REFERENCES colleges(id) ON DELETE CASCADE,
  permissions JSONB DEFAULT '{"view_students": true, "download_reports": true}',
  created_at  TIMESTAMPTZ DEFAULT NOW() NOT NULL,
  UNIQUE(user_id, college_id)
);

CREATE INDEX idx_college_admins_college ON college_admins(college_id);
CREATE INDEX idx_college_admins_user ON college_admins(user_id);

-- =====
-- TRAINERS
-- =====
CREATE TABLE trainers (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id     UUID NOT NULL UNIQUE REFERENCES users(id) ON DELETE CASCADE,
  bio         TEXT,
  specialisations TEXT[],                -- e.g., ['Python', 'ML', 'Data...
  experience_years INTEGER,
  linkedin_url TEXT,
  github_url  TEXT,
  rating      NUMERIC(3,2),
  is_active   BOOLEAN DEFAULT true NOT NULL,
  created_at  TIMESTAMPTZ DEFAULT NOW() NOT NULL,
  updated_at  TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

-- =====
-- STUDENTS
-- =====
CREATE TABLE students (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id     UUID NOT NULL UNIQUE REFERENCES users(id) ON DELETE CASCADE,
  student_code VARCHAR(20) UNIQUE NOT NULL,    -- KTC-2024-00001
  college_id  UUID REFERENCES colleges(id),
  enrollment_number VARCHAR(100),
  branch      VARCHAR(100),
  graduation_year INTEGER,
  cgpa        NUMERIC(4,2),
  gender       VARCHAR(20),
  date_of_birth DATE,
  resume_url  TEXT,
  linkedin_url TEXT,
  github_url  TEXT,
  skills       TEXT[],
  placement_status VARCHAR(50) DEFAULT 'NOT_STARTED'
  CHECK (placement_status IN (
    'NOT_STARTED', 'PREPARING', 'ACTIVELY_APPLYING',
    'PLACED', 'ON_HOLD', 'NOT_INTERESTED'
  )),
  profile_completed BOOLEAN DEFAULT false NOT NULL,
  created_at  TIMESTAMPTZ DEFAULT NOW() NOT NULL,
  updated_at  TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

```

```

CREATE INDEX idx_students_college ON students(college_id);
CREATE INDEX idx_students_code ON students(student_code);
CREATE INDEX idx_students_placement ON students(placement_status);
CREATE INDEX idx_students_grad_year ON students(graduation_year);

```

```
---
```

3.3 Course & Content Tables

```

-- =====
-- COURSES
-- =====
CREATE TABLE courses (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  title       VARCHAR(255) NOT NULL,
  slug        VARCHAR(255) UNIQUE NOT NULL,
  description  TEXT,
  short_description VARCHAR(500),
  thumbnail_url TEXT,
  category    VARCHAR(100),
  difficulty   VARCHAR(20) CHECK (difficulty IN ('BEGINNER', 'INTERMEDIATE', 'ADVA...
  duration_hours INTEGER,
  status       VARCHAR(20) DEFAULT 'DRAFT'
              CHECK (status IN ('DRAFT', 'REVIEW', 'PUBLISHED', 'ARCHIVED')),
  is_sequential BOOLEAN DEFAULT true,      -- must complete modules in order
  min_attendance_pct INTEGER DEFAULT 75,   -- for certificate eligibility
  min_quiz_avg_pct  INTEGER DEFAULT 60,    -- for certificate eligibility
  version          INTEGER DEFAULT 1,
  created_by       UUID REFERENCES users(id),
  published_at     TIMESTAMPTZ,
  created_at       TIMESTAMPTZ DEFAULT NOW() NOT NULL,
  updated_at       TIMESTAMPTZ DEFAULT NOW() NOT NULL,
  deleted_at       TIMESTAMPTZ
);

CREATE INDEX idx_courses_status ON courses(status);
CREATE INDEX idx_courses_slug ON courses(slug);

-- =====
-- MODULES (Chapters within a course)
-- =====
CREATE TABLE modules (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  course_id   UUID NOT NULL REFERENCES courses(id) ON DELETE CASCADE,
  title       VARCHAR(255) NOT NULL,
  description  TEXT,
  sort_order  INTEGER NOT NULL DEFAULT 0,
  is_locked   BOOLEAN DEFAULT false,      -- locked until previous module complete
  created_at  TIMESTAMPTZ DEFAULT NOW() NOT NULL,
  updated_at  TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

CREATE INDEX idx_modules_course ON modules(course_id);
CREATE INDEX idx_modules_order ON modules(course_id, sort_order);

```

```

-- =====
-- LESSONS
-- =====
CREATE TABLE lessons (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  module_id   UUID NOT NULL REFERENCES modules(id) ON DELETE CASCADE,
  title       VARCHAR(255) NOT NULL,
  description  TEXT,
  sort_order  INTEGER NOT NULL DEFAULT 0,
  lesson_type VARCHAR(30) NOT NULL
              CHECK (lesson_type IN ('VIDEO', 'PDF', 'TEXT', 'EXTERNAL_LINK', 'MI...
  content_text TEXT,                      -- Rich text content for TEXT type
  external_url TEXT,                      -- For EXTERNAL_LINK type
  duration_minutes INTEGER,
  is_preview  BOOLEAN DEFAULT false,      -- Can be viewed without enrollment
  is_mandatory BOOLEAN DEFAULT true,
  created_at  TIMESTAMPTZ DEFAULT NOW() NOT NULL,
  updated_at  TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

CREATE INDEX idx_lessons_module ON lessons(module_id);
CREATE INDEX idx_lessons_order ON lessons(module_id, sort_order);

-- =====
-- LESSON VIDEOS
-- =====
CREATE TABLE lesson_videos (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  lesson_id   UUID NOT NULL REFERENCES lessons(id) ON DELETE CASCADE,
  source_type VARCHAR(20) NOT NULL CHECK (source_type IN ('UPLOAD', 'YOUTUBE', 'V...
  raw_url     TEXT,                      -- Original upload URL or embed URL
  cdn_url_480p TEXT,                    -- Processed HLS URL
  cdn_url_720p TEXT,
  cdn_url_1080p TEXT,
  thumbnail_url TEXT,
  duration_seconds INTEGER,
  file_size_bytes BIGINT,
  status      VARCHAR(20) DEFAULT 'PENDING'
              CHECK (status IN ('PENDING', 'PROCESSING', 'READY', 'FAILED')),
  job_id      VARCHAR(255),              -- MediaConvert job ID
  created_at  TIMESTAMPTZ DEFAULT NOW() NOT NULL,
  updated_at  TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

CREATE INDEX idx_lesson_videos_lesson ON lesson_videos(lesson_id);
CREATE INDEX idx_lesson_videos_status ON lesson_videos(status);

-- =====
-- LESSON NOTES (PDFs uploaded by trainer)
-- =====
CREATE TABLE lesson_notes (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  lesson_id   UUID NOT NULL REFERENCES lessons(id) ON DELETE CASCADE,
  title       VARCHAR(255) NOT NULL,
  file_url    TEXT NOT NULL,
  file_size_bytes BIGINT,

```

```

        sort_order      INTEGER DEFAULT 0,
        uploaded_by     UUID REFERENCES users(id),
        created_at       TIMESTAMPTZ DEFAULT NOW() NOT NULL
    );

    CREATE INDEX idx_lesson_notes_lesson ON lesson_notes(lesson_id);

---

```

3.4 Batch & Enrollment Tables

```

-- =====
-- BATCHES
-- =====
CREATE TABLE batches (
    id            UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    name          VARCHAR(255) NOT NULL,
    code          VARCHAR(50) UNIQUE NOT NULL,          -- e.g., 'PY-AI-2024-B01'
    college_id    UUID NOT NULL REFERENCES colleges(id),
    course_id     UUID NOT NULL REFERENCES courses(id),
    start_date    DATE NOT NULL,
    end_date      DATE NOT NULL,
    mode          VARCHAR(20) DEFAULT 'HYBRID'
    CHECK (mode IN ('ONLINE', 'OFFLINE', 'HYBRID')),
    schedule_days TEXT[],                             -- ['MONDAY', 'WEDNESDAY', 'FRIDAY']
    schedule_time VARCHAR(20),                         -- '18:00-20:00'
    capacity      INTEGER DEFAULT 200,
    status        VARCHAR(20) DEFAULT 'UPCOMING'
    CHECK (status IN ('UPCOMING', 'ACTIVE', 'COMPLETED', 'CANCELLED')),
    meet_link     TEXT,                                -- Zoom/GMeet URL
    notes         TEXT,
    created_by    UUID REFERENCES users(id),
    created_at    TIMESTAMPTZ DEFAULT NOW() NOT NULL,
    updated_at    TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

CREATE INDEX idx_batches_college ON batches(college_id);
CREATE INDEX idx_batches_course ON batches(course_id);
CREATE INDEX idx_batches_status ON batches(status);

-- =====
-- BATCH TRAINERS (many-to-many)
-- =====
CREATE TABLE batch_trainers (
    id            UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    batch_id      UUID NOT NULL REFERENCES batches(id) ON DELETE CASCADE,
    trainer_id    UUID NOT NULL REFERENCES trainers(id) ON DELETE CASCADE,
    is_primary    BOOLEAN DEFAULT false,
    created_at    TIMESTAMPTZ DEFAULT NOW() NOT NULL,
    UNIQUE(batch_id, trainer_id)
);

-- =====
-- BATCH STUDENTS (Enrollment)

```

```
-- =====
CREATE TABLE batch_students (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  batch_id    UUID NOT NULL REFERENCES batches(id) ON DELETE CASCADE,
  student_id  UUID NOT NULL REFERENCES students(id) ON DELETE CASCADE,
  enrolled_at TIMESTAMPTZ DEFAULT NOW() NOT NULL,
  enrolled_by UUID REFERENCES users(id),
  status      VARCHAR(20) DEFAULT 'ACTIVE'
              CHECK (status IN ('ACTIVE', 'INACTIVE', 'DROPPED', 'COMPLETED')),
  completion_pct NUMERIC(5,2) DEFAULT 0,
  last_active_at TIMESTAMPTZ,
  UNIQUE(batch_id, student_id)
);

CREATE INDEX idx_batch_students_batch ON batch_students(batch_id);
CREATE INDEX idx_batch_students_student ON batch_students(student_id);
CREATE INDEX idx_batch_students_status ON batch_students(status);

---
```

3.5 Attendance Tables

```
-- =====
-- CLASS SESSIONS
-- =====
CREATE TABLE class_sessions (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  batch_id    UUID NOT NULL REFERENCES batches(id) ON DELETE CASCADE,
  trainer_id  UUID REFERENCES trainers(id),
  session_date DATE NOT NULL,
  start_time  TIME NOT NULL,
  end_time    TIME NOT NULL,
  topic_covered TEXT,
  mode        VARCHAR(20) DEFAULT 'ONLINE'
              CHECK (mode IN ('ONLINE', 'OFFLINE', 'HYBRID')),
  recording_url TEXT,
  session_notes TEXT,
  total_students INTEGER DEFAULT 0,
  present_count  INTEGER DEFAULT 0,
  attendance_marked BOOLEAN DEFAULT false,
  attendance_marked_at TIMESTAMPTZ,
  created_at    TIMESTAMPTZ DEFAULT NOW() NOT NULL,
  updated_at    TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

CREATE INDEX idx_sessions_batch ON class_sessions(batch_id);
CREATE INDEX idx_sessions_date ON class_sessions(session_date);
CREATE INDEX idx_sessions_trainer ON class_sessions(trainer_id);
```

```
-- =====
-- ATTENDANCE RECORDS
-- =====
CREATE TABLE attendance_records (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```



```

    session_id      UUID NOT NULL REFERENCES class_sessions(id) ON DELETE CASCADE,
    student_id      UUID NOT NULL REFERENCES students(id) ON DELETE CASCADE,
    status          VARCHAR(20) NOT NULL CHECK (status IN ('PRESENT', 'ABSENT', 'LATE',...
    marked_by       UUID REFERENCES users(id),
    override_reason TEXT,                                -- Set if super admin overrides
    created_at      TIMESTAMPTZ DEFAULT NOW() NOT NULL,
    updated_at      TIMESTAMPTZ DEFAULT NOW() NOT NULL,
    UNIQUE(session_id, student_id)
);

CREATE INDEX idx_attendance_session ON attendance_records(session_id);
CREATE INDEX idx_attendance_student ON attendance_records(student_id);
CREATE INDEX idx_attendance_status ON attendance_records(status);

-- =====
-- ATTENDANCE SUMMARY (Materialised for performance)
-- =====
CREATE MATERIALIZED VIEW attendance_summary AS
SELECT
    bs.batch_id,
    ar.student_id,
    COUNT(*) FILTER (WHERE cs.attendance_marked = true) AS total_sessions,
    COUNT(*) FILTER (WHERE ar.status IN ('PRESENT', 'LATE')) AS attended_sessions,
    ROUND(
        COUNT(*) FILTER (WHERE ar.status IN ('PRESENT', 'LATE')) * 100.0 /
        NULLIF(COUNT(*) FILTER (WHERE cs.attendance_marked = true), 0), 2
    ) AS attendance_pct
FROM batch_students bs
LEFT JOIN class_sessions cs ON cs.batch_id = bs.batch_id
LEFT JOIN attendance_records ar ON ar.session_id = cs.id AND ar.student_id = bs.student_id
GROUP BY bs.batch_id, ar.student_id;

CREATE INDEX idx_attendance_summary_batch ON attendance_summary(batch_id);
CREATE INDEX idx_attendance_summary_student ON attendance_summary(student_id);

-- Refresh this view nightly and after attendance marking

---
```

3.6 Assessment Tables

```

-- =====
-- QUIZZES
-- =====
CREATE TABLE quizzes (
    id              UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    title           VARCHAR(255) NOT NULL,
    description      TEXT,
    course_id       UUID REFERENCES courses(id),
    module_id       UUID REFERENCES modules(id),
    batch_id        UUID REFERENCES batches(id),          -- if batch-specific
    created_by      UUID REFERENCES users(id),
    time_limit_mins INTEGER,                             -- NULL = no time limit
    total_marks     INTEGER NOT NULL,
```

```

    pass_marks          INTEGER NOT NULL,
    attempts_allowed    INTEGER DEFAULT 1,
    shuffle_questions    BOOLEAN DEFAULT false,
    shuffle_options      BOOLEAN DEFAULT true,
    show_answers_immediately BOOLEAN DEFAULT true,
    available_from       TIMESTAMPTZ,
    available_until      TIMESTAMPTZ,
    status               VARCHAR(20) DEFAULT 'DRAFT'
                        CHECK (status IN ('DRAFT', 'PUBLISHED', 'CLOSED')),
    created_at           TIMESTAMPTZ DEFAULT NOW() NOT NULL,
    updated_at           TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

CREATE INDEX idx_quizzes_batch ON quizzes(batch_id);
CREATE INDEX idx_quizzes_course ON quizzes(course_id);
CREATE INDEX idx_quizzes_status ON quizzes(status);

-- =====
-- QUIZ QUESTIONS
-- =====
CREATE TABLE quiz_questions (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    quiz_id           UUID NOT NULL REFERENCES quizzes(id) ON DELETE CASCADE,
    question_text      TEXT NOT NULL,
    question_type      VARCHAR(30) NOT NULL
                    CHECK (question_type IN ('MCQ_SINGLE', 'MCQ_MULTIPLE', 'TRUE_FALSE'...
    marks              INTEGER NOT NULL DEFAULT 1,
    difficulty         VARCHAR(20) DEFAULT 'MEDIUM'
                    CHECK (difficulty IN ('EASY', 'MEDIUM', 'HARD')),
    topic_tag          VARCHAR(100),
    explanation        TEXT,                                -- Shown after answering
    sort_order         INTEGER DEFAULT 0,
    created_at         TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

CREATE INDEX idx_quiz_questions_quiz ON quiz_questions(quiz_id);

-- =====
-- QUESTION OPTIONS
-- =====
CREATE TABLE question_options (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    question_id       UUID NOT NULL REFERENCES quiz_questions(id) ON DELETE CASCADE,
    option_text       TEXT NOT NULL,
    is_correct        BOOLEAN NOT NULL DEFAULT false,
    sort_order        INTEGER DEFAULT 0
);

CREATE INDEX idx_question_options_question ON question_options(question_id);

-- =====
-- QUIZ ATTEMPTS
-- =====
CREATE TABLE quiz_attempts (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    quiz_id           UUID NOT NULL REFERENCES quizzes(id) ON DELETE CASCADE,

```

```

    student_id      UUID NOT NULL REFERENCES students(id) ON DELETE CASCADE,
    attempt_number  INTEGER NOT NULL DEFAULT 1,
    started_at      TIMESTAMPTZ DEFAULT NOW() NOT NULL,
    submitted_at    TIMESTAMPTZ,
    time_taken_secs INTEGER,
    score           INTEGER,
    max_score       INTEGER,
    percentage      NUMERIC(5,2),
    passed          BOOLEAN,
    status          VARCHAR(20) DEFAULT 'IN_PROGRESS'
                  CHECK (status IN ('IN_PROGRESS', 'SUBMITTED', 'TIMED_OUT', 'ABANDON...
    ip_address      INET,
    last_saved_at   TIMESTAMPTZ                -- auto-save checkpoint
);

CREATE INDEX idx_quiz_attempts_quiz ON quiz_attempts(quiz_id);
CREATE INDEX idx_quiz_attempts_student ON quiz_attempts(student_id);
CREATE INDEX idx_quiz_attempts_status ON quiz_attempts(status);

-- =====
-- QUIZ ANSWERS
-- =====
CREATE TABLE quiz_answers (
    id              UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    attempt_id      UUID NOT NULL REFERENCES quiz_attempts(id) ON DELETE CASCADE,
    question_id     UUID NOT NULL REFERENCES quiz_questions(id),
    selected_option_ids UUID[],                -- For MCQ types
    text_answer     TEXT,                     -- For SHORT_TEXT type
    is_correct      BOOLEAN,
    marks_awarded   NUMERIC(5,2) DEFAULT 0,
    answered_at     TIMESTAMPTZ DEFAULT NOW(),
    UNIQUE(attempt_id, question_id)
);

CREATE INDEX idx_quiz_answers_attempt ON quiz_answers(attempt_id);

-- =====
-- ASSIGNMENTS
-- =====
CREATE TABLE assignments (
    id              UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    title           VARCHAR(255) NOT NULL,
    description     TEXT,
    course_id       UUID REFERENCES courses(id),
    module_id       UUID REFERENCES modules(id),
    batch_id        UUID REFERENCES batches(id),
    created_by      UUID REFERENCES users(id),
    total_marks     INTEGER NOT NULL,
    pass_marks      INTEGER,
    deadline_at     TIMESTAMPTZ,
    allow_late      BOOLEAN DEFAULT true,
    submission_type VARCHAR(20) DEFAULT 'FILE'
                  CHECK (submission_type IN ('FILE', 'TEXT', 'LINK', 'MIXED')),
    instructions    TEXT,
    resource_url    TEXT,                    -- Reference material
    status          VARCHAR(20) DEFAULT 'DRAFT'

```

```

        CHECK (status IN ('DRAFT', 'PUBLISHED', 'CLOSED')),
    created_at    TIMESTAMPTZ DEFAULT NOW() NOT NULL,
    updated_at    TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

CREATE INDEX idx_assignments_batch ON assignments(batch_id);
CREATE INDEX idx_assignments_deadline ON assignments(deadline_at);

-- =====
-- ASSIGNMENT SUBMISSIONS
-- =====
CREATE TABLE assignment_submissions (
    id            UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    assignment_id  UUID NOT NULL REFERENCES assignments(id) ON DELETE CASCADE,
    student_id    UUID NOT NULL REFERENCES students(id) ON DELETE CASCADE,
    submission_text TEXT,
    file_url      TEXT,
    submission_link TEXT,
    student_note  TEXT,
    submitted_at  TIMESTAMPTZ DEFAULT NOW() NOT NULL,
    is_late       BOOLEAN DEFAULT false,
    marks_awarded NUMERIC(5,2),
    feedback      TEXT,
    graded_by     UUID REFERENCES users(id),
    graded_at     TIMESTAMPTZ,
    status        VARCHAR(20) DEFAULT 'SUBMITTED'
                CHECK (status IN ('SUBMITTED', 'UNDER_REVIEW', 'GRADED', 'RETURNED')),
    UNIQUE(assignment_id, student_id)
);

CREATE INDEX idx_submissions_assignment ON assignment_submissions(assignment_id);
CREATE INDEX idx_submissions_student ON assignment_submissions(student_id);
CREATE INDEX idx_submissions_status ON assignment_submissions(status);

---
```

3.7 Progress & Analytics Tables

```

-- =====
-- STUDENT LESSON PROGRESS
-- =====
CREATE TABLE student_progress (
    id            UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    student_id    UUID NOT NULL REFERENCES students(id) ON DELETE CASCADE,
    lesson_id     UUID NOT NULL REFERENCES lessons(id) ON DELETE CASCADE,
    batch_id      UUID REFERENCES batches(id),
    started_at    TIMESTAMPTZ DEFAULT NOW(),
    completed_at  TIMESTAMPTZ,
    video_progress_secs INTEGER DEFAULT 0,           -- Last watched position
    video_completed_pct NUMERIC(5,2) DEFAULT 0,
    is_completed  BOOLEAN DEFAULT false,
    UNIQUE(student_id, lesson_id)
);

CREATE INDEX idx_progress_student ON student_progress(student_id);
```

```

CREATE INDEX idx_progress_lesson ON student_progress(lesson_id);
CREATE INDEX idx_progress_completed ON student_progress(is_completed);

-- =====
-- STUDENT NOTES (Personal notes on lessons)
-- =====
CREATE TABLE student_notes (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    student_id        UUID NOT NULL REFERENCES students(id) ON DELETE CASCADE,
    lesson_id         UUID NOT NULL REFERENCES lessons(id) ON DELETE CASCADE,
    content           TEXT NOT NULL,
    created_at        TIMESTAMPTZ DEFAULT NOW() NOT NULL,
    updated_at        TIMESTAMPTZ DEFAULT NOW() NOT NULL,
    UNIQUE(student_id, lesson_id)
);

-- =====
-- ACTIVITY LOG (for streak calculation and heatmap)
-- =====
CREATE TABLE student_activity_log (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    student_id        UUID NOT NULL REFERENCES students(id) ON DELETE CASCADE,
    activity_type      VARCHAR(50) NOT NULL
                        CHECK (activity_type IN (
                            'LESSON_VIEWED', 'VIDEO_WATCHED', 'QUIZ_ATTEMPTED',
                            'ASSIGNMENT_SUBMITTED', 'NOTE_WRITTEN', 'LOGIN'
                        )),
    entity_id         UUID, -- ID of related entity
    duration_secs     INTEGER,
    activity_date      DATE NOT NULL DEFAULT CURRENT_DATE,
    created_at        TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

CREATE INDEX idx_activity_student ON student_activity_log(student_id);
CREATE INDEX idx_activity_date ON student_activity_log(student_id, activity_date);

---
```

3.8 Certificate & Document Tables

```

-- =====
-- CERTIFICATES
-- =====
CREATE TABLE certificates (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    certificate_code   VARCHAR(20) UNIQUE NOT NULL, -- KTC-CERT-2024-XXXXX
    student_id        UUID NOT NULL REFERENCES students(id),
    course_id         UUID NOT NULL REFERENCES courses(id),
    batch_id          UUID REFERENCES batches(id),
    issued_at         TIMESTAMPTZ DEFAULT NOW() NOT NULL,
    issued_by         UUID REFERENCES users(id),
    pdf_url           TEXT NOT NULL,
    is_valid           BOOLEAN DEFAULT true,
    revoked_at        TIMESTAMPTZ,

```

```

        revoke_reason    TEXT,
        metadata         JSONB DEFAULT '{}',                -- trainer name, etc.
        created_at       TIMESTAMPTZ DEFAULT NOW() NOT NULL
    );

CREATE INDEX idx_certificates_student ON certificates(student_id);
CREATE INDEX idx_certificates_code ON certificates(certificate_code);
CREATE INDEX idx_certificates_course ON certificates(course_id);

-- =====
-- STUDENT DOCUMENTS (Offer letters, internship letters)
-- =====
CREATE TABLE student_documents (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    student_id        UUID NOT NULL REFERENCES students(id) ON DELETE CASCADE,
    document_type      VARCHAR(50) NOT NULL
                        CHECK (document_type IN (
                            'OFFER_LETTER', 'INTERNSHIP_LETTER', 'COMPLETION_LETTER',
                            'RECOMMENDATION_LETTER', 'OTHER'
                        )),
    title             VARCHAR(255) NOT NULL,
    file_url          TEXT NOT NULL,
    company_name       VARCHAR(255),
    issue_date        DATE,
    uploaded_by       UUID REFERENCES users(id),
    notes             TEXT,
    created_at        TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

CREATE INDEX idx_documents_student ON student_documents(student_id);
CREATE INDEX idx_documents_type ON student_documents(document_type);

```

3.9 Placement Tables

```

-- =====
-- PLACEMENT RECORDS
-- =====
CREATE TABLE placement_records (
    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    student_id        UUID NOT NULL REFERENCES students(id) ON DELETE CASCADE,
    company_name       VARCHAR(255) NOT NULL,
    role_title        VARCHAR(255) NOT NULL,
    offer_type        VARCHAR(30) NOT NULL
                        CHECK (offer_type IN ('FULL_TIME', 'INTERNSHIP', 'CONTRACT', 'FREEL...
    ctc_annual        NUMERIC(12,2),                        -- Annual CTC in INR
    stipend_monthly    NUMERIC(10,2),                        -- For internships
    offer_date        DATE,
    joining_date      DATE,
    location          VARCHAR(255),
    is_verified       BOOLEAN DEFAULT false,
    verified_by       UUID REFERENCES users(id),
    document_id       UUID REFERENCES student_documents(id),

```

```

    source          VARCHAR(50) DEFAULT 'COLLEGE_DRIVE'
                   CHECK (source IN ('COLLEGE_DRIVE', 'JOB_BOARD', 'REFERRAL', 'SELF',...
    notes           TEXT,
    created_by      UUID REFERENCES users(id),
    created_at      TIMESTAMPTZ DEFAULT NOW() NOT NULL,
    updated_at      TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

```

```

CREATE INDEX idx_placement_student ON placement_records(student_id);
CREATE INDEX idx_placement_company ON placement_records(company_name);
CREATE INDEX idx_placement_date ON placement_records(offer_date);

```

```

-- =====
-- JOB OPPORTUNITIES
-- =====
CREATE TABLE job_opportunities (
    id              UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    title           VARCHAR(255) NOT NULL,
    company_name    VARCHAR(255) NOT NULL,
    company_logo_url TEXT,
    job_type        VARCHAR(30) NOT NULL
                   CHECK (job_type IN ('FULL_TIME', 'INTERNSHIP', 'CONTRACT')),
    description     TEXT,
    skills_required TEXT[],
    ctc_min         NUMERIC(12,2),
    ctc_max         NUMERIC(12,2),
    stipend_monthly NUMERIC(10,2),
    location        VARCHAR(255),
    is_remote       BOOLEAN DEFAULT false,
    application_link TEXT,
    application_email VARCHAR(255),
    apply_deadline  DATE,
    target_batches  UUID[],                      -- specific batch IDs or null f...
    target_colleges UUID[],                      -- specific college IDs or null...
    is_active       BOOLEAN DEFAULT true,
    posted_by      UUID REFERENCES users(id),
    created_at      TIMESTAMPTZ DEFAULT NOW() NOT NULL,
    updated_at      TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

```

```

CREATE INDEX idx_jobs_active ON job_opportunities(is_active) WHERE is_active = true;
CREATE INDEX idx_jobs_deadline ON job_opportunities(apply_deadline);

```

```

-- =====
-- JOB INTERESTS (Student expresses interest in a job)
-- =====
CREATE TABLE job_interests (
    id              UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    job_id          UUID NOT NULL REFERENCES job_opportunities(id) ON DELETE CASCADE,
    student_id      UUID NOT NULL REFERENCES students(id) ON DELETE CASCADE,
    status          VARCHAR(30) DEFAULT 'INTERESTED'
                   CHECK (status IN ('INTERESTED', 'APPLIED', 'SHORTLISTED', 'REJECTED', '...
    created_at      TIMESTAMPTZ DEFAULT NOW() NOT NULL,
    UNIQUE(job_id, student_id)
);

```

```
CREATE INDEX idx_job_interests_job ON job_interests(job_id);
CREATE INDEX idx_job_interests_student ON job_interests(student_id);
```

```
---
```

3.10 Communication Tables

```
-- =====
-- NOTIFICATIONS
-- =====
CREATE TABLE notifications (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  type        VARCHAR(50) NOT NULL,          -- 'ANNOUNCEMENT', 'SYSTEM', 'Q...'
  title       VARCHAR(255) NOT NULL,
  body        TEXT NOT NULL,
  data        JSONB DEFAULT '{}',           -- Deep link data
  created_by  UUID REFERENCES users(id),
  target_scope VARCHAR(30) NOT NULL
              CHECK (target_scope IN ('ALL', 'COLLEGE', 'BATCH', 'INDIVIDUAL')),
  target_id   UUID,                         -- college_id, batch_id, or use...
  created_at  TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

CREATE INDEX idx_notifications_scope ON notifications(target_scope, target_id);
CREATE INDEX idx_notifications_created ON notifications(created_at DESC);

-- =====
-- NOTIFICATION RECIPIENTS (individual delivery records)
-- =====
CREATE TABLE notification_recipients (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  notification_id UUID NOT NULL REFERENCES notifications(id) ON DELETE CASCADE,
  user_id     UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
  is_read     BOOLEAN DEFAULT false,
  read_at     TIMESTAMPTZ,
  push_sent   BOOLEAN DEFAULT false,
  email_sent  BOOLEAN DEFAULT false,
  created_at  TIMESTAMPTZ DEFAULT NOW() NOT NULL,
  UNIQUE(notification_id, user_id)
);

CREATE INDEX idx_notif_recipients_user ON notification_recipients(user_id);
CREATE INDEX idx_notif_recipients_unread ON notification_recipients(user_id, is_read)
  WHERE is_read = false;
```

```
---
```

3.11 System Tables

```
-- =====
-- AUDIT LOGS (All admin actions)
-- =====
CREATE TABLE audit_logs (
```



```

    id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id            UUID REFERENCES users(id),
    action              VARCHAR(100) NOT NULL,          -- 'CREATE', 'UPDATE', 'DELETE...'
    entity_type         VARCHAR(100) NOT NULL,
    entity_id           UUID,
    old_values           JSONB,
    new_values           JSONB,
    ip_address           INET,
    user_agent           TEXT,
    created_at           TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

CREATE INDEX idx_audit_user ON audit_logs(user_id);
CREATE INDEX idx_audit_entity ON audit_logs(entity_type, entity_id);
CREATE INDEX idx_audit_created ON audit_logs(created_at DESC);

-- =====
-- SYSTEM SETTINGS
-- =====
CREATE TABLE system_settings (
    key                VARCHAR(100) PRIMARY KEY,
    value              TEXT NOT NULL,
    description         TEXT,
    updated_at          TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

-- Seed data
INSERT INTO system_settings VALUES
    ('default_attendance_threshold', '75', 'Minimum attendance % for certificate eligib...',
    ('default_quiz_pass_pct', '60', 'Minimum quiz average % for certificate eligibility'),
    ('student_code_prefix', 'KTC', 'Prefix for student codes'),
    ('certificate_code_prefix', 'KTC-CERT', 'Prefix for certificate codes'),
    ('max_file_upload_mb', '50', 'Maximum PDF upload size in MB'),
    ('max_video_upload_gb', '2', 'Maximum video upload size in GB');

---
```

4. Key Indexes Summary

```

-- Performance indexes for common query patterns
CREATE INDEX idx_batch_students_active ON batch_students(batch_id, student_id)
    WHERE status = 'ACTIVE';

CREATE INDEX idx_sessions_unmarked ON class_sessions(batch_id, session_date)
    WHERE attendance_marked = false;

CREATE INDEX idx_quiz_attempts_incomplete ON quiz_attempts(quiz_id, student_id)
    WHERE status = 'IN_PROGRESS';

CREATE INDEX idx_students_at_risk ON batch_students(batch_id)
    WHERE completion_pct < 0.4;

---
```

5. Auto-Update Timestamps Trigger

```
CREATE OR REPLACE FUNCTION update_updated_at()  
RETURNS TRIGGER AS $$  
BEGIN  
    NEW.updated_at = NOW();  
    RETURN NEW;  
END;  
$$ LANGUAGE plpgsql;  
  
-- Apply to all tables with updated_at  
CREATE TRIGGER trg_users_updated_at BEFORE UPDATE ON users  
    FOR EACH ROW EXECUTE FUNCTION update_updated_at();  
CREATE TRIGGER trg_colleges_updated_at BEFORE UPDATE ON colleges  
    FOR EACH ROW EXECUTE FUNCTION update_updated_at();  
CREATE TRIGGER trg_courses_updated_at BEFORE UPDATE ON courses  
    FOR EACH ROW EXECUTE FUNCTION update_updated_at();  
-- (repeat for all relevant tables)
```

End of Database Schema Document